NixieSync r5 — drawing set

Contents

Board specification

Size	50.0 × 30.0 mm, 2.0 mm corner radius
Stackup	2 layer, 1.6 mm FR4, 1 oz copper
Finish	ENIG preferred for flatness; HASL acceptable
Min track / gap	0.20 mm / 0.20 mm
Default track	0.25 mm signal, 0.50 mm +5V
Vias	0.30 mm drill / 0.60 mm pad
Bottom layer	continuous GND pour, stitched every ~10 mm
Top layer	signal plus a +5V pour
Mounting	4 × M2.5, Ø2.7 mm, 5.5 mm keep-out
Max height	~3.5 mm (the XIAO's USB-C shell); allow 8 mm
Power in	5 V via the XIAO's own USB-C, overhanging the left edge

Reset method

Q3 pulls the ST62's RESET pin low for ~50 ms. Confirmed on the hardware: grounding that pin through 1 kΩ returns the display to 00:00:00, the same known state a power cycle gives.

The net is already a wired-OR node — an MC33064-5 undervoltage supervisor (open collector, ~4.6 V threshold, confirmed) and the MCU's own watchdog and power-on reset all pull it low. Q3 simply joins that drive, so nothing about the clock's own reset behaviour changes.

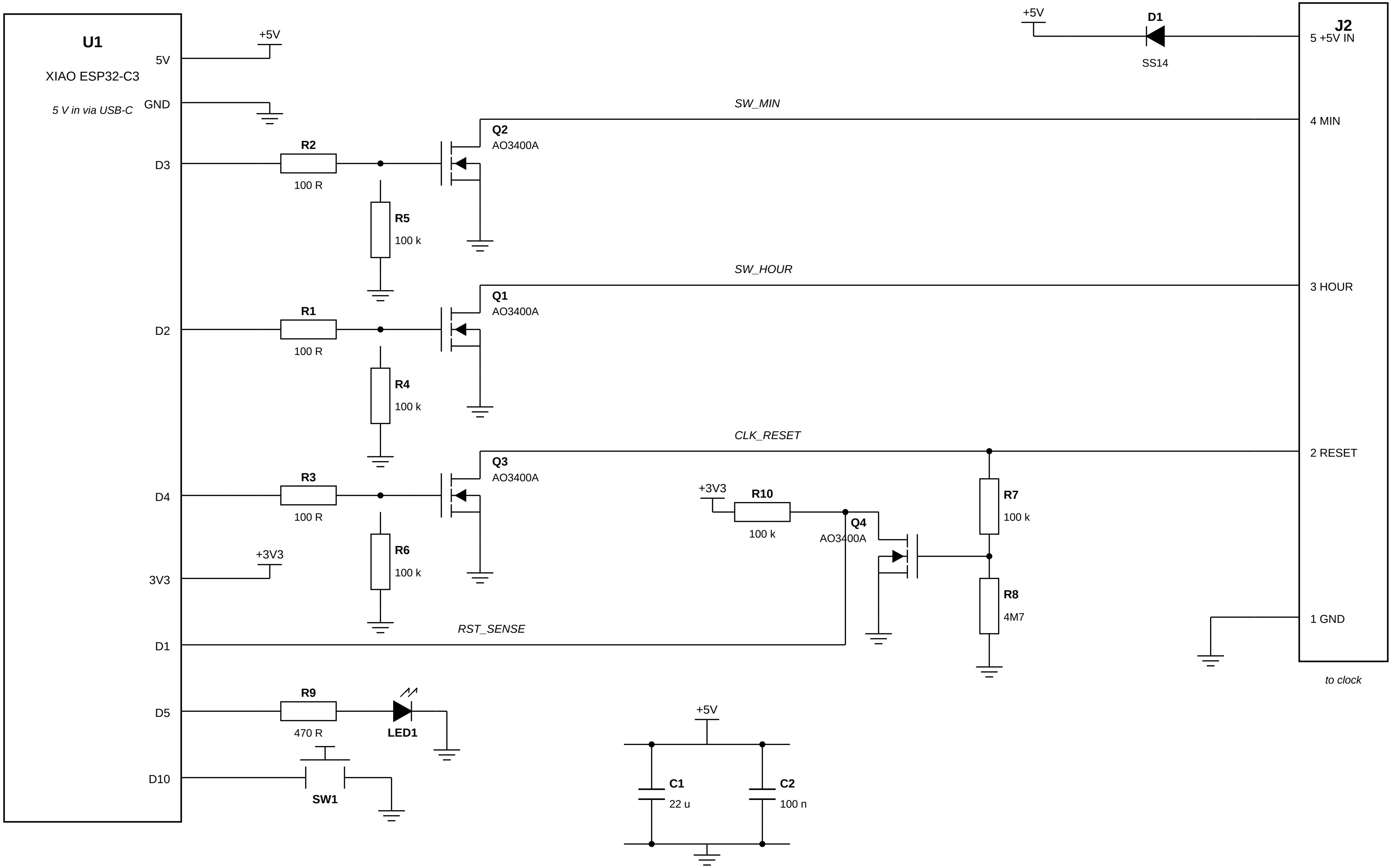
Q4 buffers the same node for sensing, presenting only R7+R8 of gate pull-down instead of a divider. That matters: this is the one net on the board whose loading could hold the clock in reset permanently, so the sense path is deliberately kept out of its DC operating point.

An earlier revision carried a relay to break the ST62's VDD as a fallback. The reset test closed that question, so K3, D3 and both selection jumpers are gone and J2 is now 5-way.

NixieSync r5 — circuit diagram

All four outputs are low-side N-FETs. The clock pulls MIN, HOUR and RESET up through 1 kohm; we pull them down.

The switch common is the clock's ground, so no separate COMMON wire is needed — J2-1 serves as both.   AO3400A throughout: a 2N7002 is not reliably enhanced by a 3.3 V gate.   D1 lets the clock's 5 V rail and a USB-C programming lead coexist.

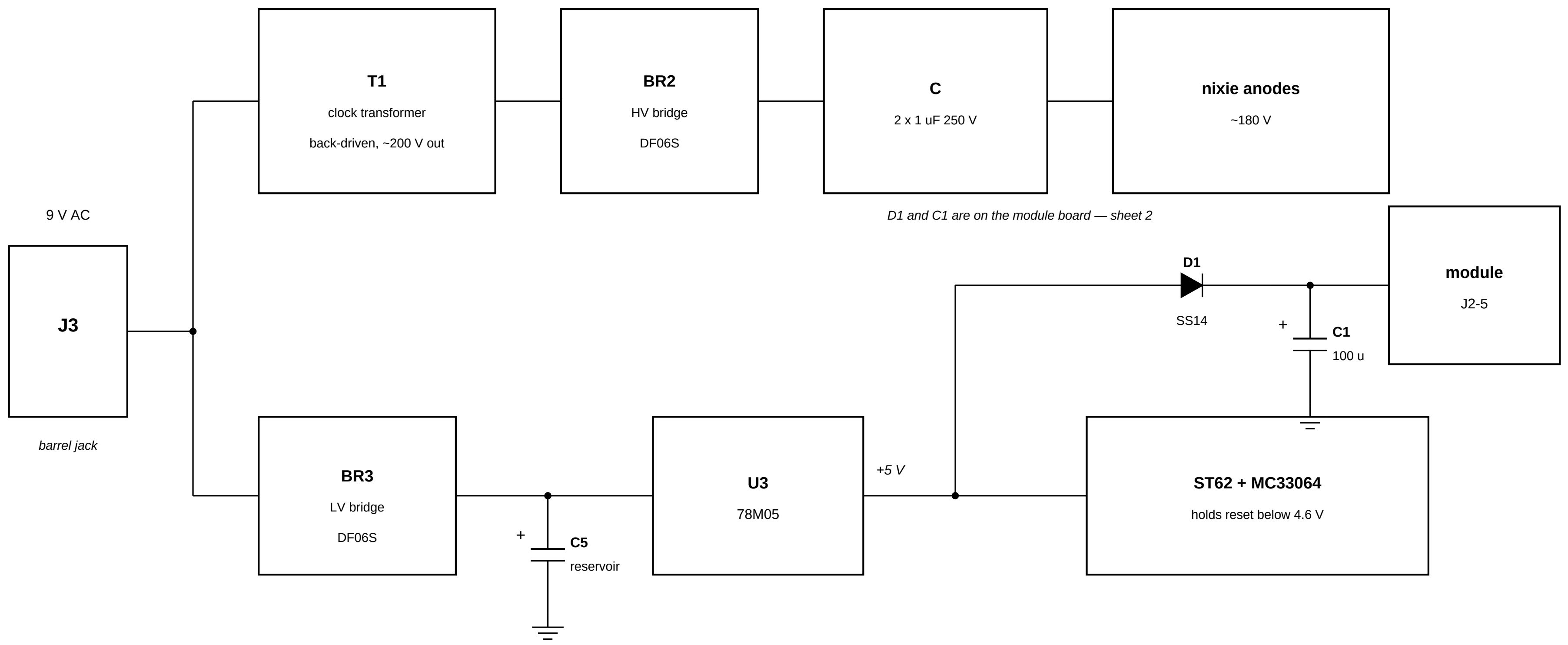


NTP time-set module for an ST62T65C6 nixie clock · 50.0 × 30.0 mm · 2 layer, 1.6 mm FR4
No relays and no switched inductive current: the only load is ~5 mA into the clock's 1 kohm pull-ups.

NixieSync r5 — power — tapping the clock's 5 V rail

The module is fed from the clock's existing 78M05 rather than from a second rectifier. Read the note below right before wiring it.

The barrel jack feeds two paths: the clock's transformer back-driven to make the ~180 V anode supply, and a separate low-voltage bridge into the 78M05.



Loading the 78M05 — check three things

1. Dissipation is (Vin - 5) x I. Measure Vin at the 78M05 input. At 11 V and 90 mA average that is 0.5 W; a lightly loaded "9 V" unit can reach 12 V AC, giving 15 V DC and 0.9 W. Ambient inside a sealed chassis warmed by six tubes is not 25 C.
2. Measure the regulator tab temperature before and after adding the load.
3. Check the ripple trough at the 78M05 input stays above ~7 V with the extra current drawn from C5 every cycle.

The trap: this rail is what the supervisor watches

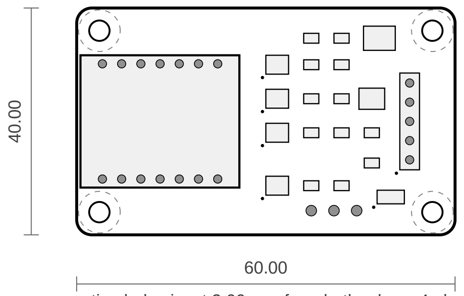
The MC33064 holds the ST62 in reset below ~4.6 V. WiFi transmit peaks are a few hundred mA for a few hundred microseconds, drawn straight through the XIAO's linear LDO from this same rail. If a peak dips it past the threshold, the clock resets — and the module then sees a foreign reset and re-syncs it, so the fault self-heals and looks like the clock occasionally resetting for no reason.

The instrument is already built in: the firmware logs "[sense] foreign reset seen" on serial every time. Watch it for a few days after wiring this up.

Mitigations: fit C1 at 100 uF so peaks come from local bulk; set WiFi.setSleep(true) to roughly halve the average draw; keep the 5 V feed short and thick.

NixieSync r5 — mechanical outline and drill plan

origin: lower-left corner, X right, Y up



mounting holes inset 3.00 mm from both edges, 4 places

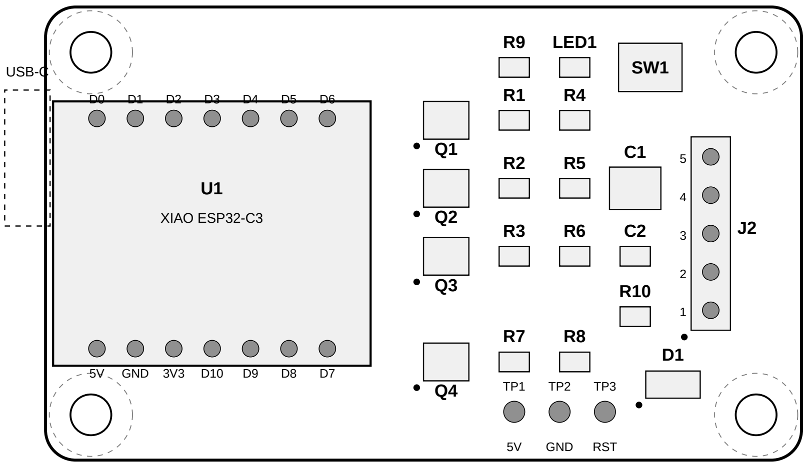
Drill table

Ref	X (mm)	Y (mm)	Ø (mm)
H1	3.00	3.00	2.70
H2	3.00	27.00	2.70
H3	47.00	3.00	2.70
H4	47.00	27.00	2.70

Print this sheet at 100 % and measure the 60.00 mm dimension before trusting it. Cut it out and test-fit against the chassis; the clock interior measured about 40 mm across in your photographs, which is what set the board width.

Component outlines are shown for interference checking only — they are body envelopes, not footprints. Nothing on the board is taller than the XIAO module itself. The XIAO's USB-C connector overhangs the left board edge so the cable can be plugged in with the board mounted.

TOP VIEW — silkscreen and body outlines



J2 to clock

1 GND
2 RESET
3 HOUR
4 MIN
5 +5V IN

5-way harness; take GND at the switch common terminal. Pin 5 optional - USB-C powers it too

Routing notes

1. Q4 buffers the clock's RESET net so the sense path loads it with ~4.8 MΩ of gate pull-down instead of a divider. Keep R7/R8 beside Q4's gate, not beside J2.
2. R4, R5 and R6 go beside their FET gates, not beside the XIAO. They hold the outputs off during the ~200 ms before setup() runs; a long open gate trace defeats that.
3. SW_HOUR, SW_MIN and CLK_RESET each sink ~5 mA into the clock's 1 kΩ pull-ups, so 0.25 mm tracks are ample. Keep them away from the XIAO's antenna end.
4. Take J2-1 (GND) to the switch common terminal itself. That is the node the clock's inputs are referenced to, and it gives the FET sources the shortest return path.
5. The XIAO ESP32-C3 uses an external u.FL antenna and the chassis is metal. Route the pigtail out of the box or seat it against a non-metallic part of the housing.
6. Mount on nylon standoffs unless you deliberately want the chassis bonded to GND.

NixieSync r5 — placement and netlist

Placement (part centre, mm, origin lower-left)

Ref	X	Y	Body W×H	Part
U1	11.00	15.00	21.0×17.5	XIAO ESP32-C3
Q1	26.50	22.50	3.0×2.5	AO3400A SOT-23 (HOUR)
Q2	26.50	18.00	3.0×2.5	AO3400A SOT-23 (MIN)
Q3	26.50	13.50	3.0×2.5	AO3400A SOT-23 (RESET)
Q4	26.50	6.50	3.0×2.5	AO3400A SOT-23 (sense buffer)
R1	31.00	22.50	2.0×1.3	100 Ω 0805
R2	31.00	18.00	2.0×1.3	100 Ω 0805
R3	31.00	13.50	2.0×1.3	100 Ω 0805
R7	31.00	6.50	2.0×1.3	100 kΩ 0805 (Q4 gate series)
R4	35.00	22.50	2.0×1.3	100 kΩ 0805
R5	35.00	18.00	2.0×1.3	100 kΩ 0805
R6	35.00	13.50	2.0×1.3	100 kΩ 0805
R8	35.00	6.50	2.0×1.3	4.7 MΩ 0805 (Q4 gate pull-down)
C1	39.00	18.00	3.4×2.8	47 uF 16 V X5R 1210
C2	39.00	13.50	2.0×1.3	100 nF 0805
R10	39.00	9.50	2.0×1.3	100 kΩ 0805 (Q4 drain pull-up)
R9	31.00	26.00	2.0×1.3	470 Ω 0805
LED1	35.00	26.00	2.0×1.3	LED 0805
SW1	40.00	26.00	4.2×3.2	SMD tact 4.2 × 3.2
D1	41.50	5.00	3.6×1.8	SS14 SOD-123 (5 V input OR-ing)
J2	44.00	15.00	2.6×12.8	5-way 2.54 mm right-angle header

Netlist

```
+5V_IN      J2.5    D1.A
+5V         D1.K    U1.5V  C1.1   C2.1  TP1
+3V3        U1.3V3  R10.1
GND         U1.GND  C1.2   C2.2   R4.2  R5.2  R6.2  R8.2
            Q1.S    Q2.S   Q3.S   Q4.S  SW1.2  LED1.K  J2.1  TP2
HOUR_DRV    U1.D2   R1.1
HOUR_GATE   R1.2    Q1.G   R4.1
SW_HOUR     Q1.D    J2.3
MIN_DRV     U1.D3   R2.1
MIN_GATE    R2.2    Q2.G   R5.1
SW_MIN      Q2.D    J2.4
RST_DRV     U1.D4   R3.1
RST_GATE    R3.2    Q3.G   R6.1
CLK_RESET   Q3.D    R7.1   J2.2   TP3
SENSE_GATE  R7.2    R8.1   Q4.G
RST_SENSE   Q4.D    R10.2  U1.D1
LED_DRV     U1.D5   R9.1
LED_A       R9.2    LED1.A
BTN         U1.D10  SW1.1
```

Unused on U1: D0, D6, D7, D8, D9. Every driver is a low-side FET — see sheet 7.

NixieSync r5 — fabrication, assembly and BOM

Why AO3400A and not 2N7002

All four gates are driven from XIAO GPIOs at 3.3 V. The 2N7002's threshold runs as high as 3.0 V and its on-resistance is specified at Vgs = 10 V, so a worst-case part is barely enhanced at 3.3 V — it would probably work and might not, which is the worst kind of part choice. The AO3400A has a 0.4–1.1 V threshold and is fully specified at 2.5 V. Any logic-level equivalent (BSS138, DMN2075U) is fine; a plain 2N7002 is not.

Q4's gate is driven from the clock's 5 V reset line rather than a GPIO, so it would tolerate a 2N7002, but using one part for all four keeps the BOM to a single line.

Earlier revisions used relays for HOUR and MIN and warned at length about dry-circuit contact wear. Both are moot: the clock's 1 kΩ pull-ups and grounded switch common mean a low-side FET reproduces a switch closure exactly, sinking about 5 mA. That also removes the only switched inductive current on the board, a few centimetres from a 200 V boost converter.

Assembly sequence

1. Passives and capacitors, then the four FETs.
2. J2, then SW1 and LED1.
3. Before fitting U1, power the bare board from TP1/TP2 and check that the Q1–Q3 drains float (gates held down by R4–R6) and that TP3 is not being pulled low.
4. U1 last, so the module is not reflowed twice.

Bill of materials

Qty	Refs	Part	Note
1	U1	Seeed XIAO ESP32-C3	external u.FL antenna
4	Q1–Q4	AO3400A SOT-23	logic level; see the note above
3	R1–R3	100 Ω 0805	gate series
3	R4–R6	100 kΩ 0805	gate pull-down, not optional
1	R7	100 kΩ 0805	Q4 gate series
1	R8	4.7 MΩ 0805	Q4 gate pull-down
1	R10	100 kΩ 0805	Q4 drain pull-up to 3V3
1	R9	470 Ω 0805	~3 mA with a red LED
1	D1	SS14 SOD-123	blocks back-feed into USB VBUS
1	C1	47 uF 16 V X5R 1210	100 uF if fed from the clock
1	C2	100 nF 0805	decoupling
1	LED1	LED 0805, RED or amber	see the note below
1	SW1	SMD tact 4.2 × 3.2 mm	resync / config wipe
1	J2	5-way 2.54 mm right-angle header	to the clock
4	H1–H4	M2.5 nylon standoff	isolate from chassis

Open items before ordering

- CONFIRMED: the SO-8 is an MC33064-5 and the clock's VDD is 5 V. Q4 buffers the sense path, so no 5 V reaches U1.D1 and the divider ratio no longer matters.
- Confirm the ST62 RESET pin. Figure 2 of the ST6255C/ST6265C datasheet reads as pin 23, next to OSCout at 22, which the crystal identifies for you. It should idle at VDD.
- Buzz out the six switch wires: hour, minute, shared common.
- CONFIRMED: 1 kohm pull-ups to VDD, switch common at ground. Hence low-side FETs throughout and a 4-way harness.
- LED1 must be red or amber (Vf ~1.9 V). The GPIO drives it from 3.3 V, so a green or blue part at Vf ~3.0 V has only a couple of hundred mV of headroom: barely lit, and the current swings with tolerance and temperature. R9 = 470 ohm gives ~3 mA with a red LED.
- Pull the XIAO footprint from Seeed. The envelopes in this drawing set are for clearance checking only. Everything else is a standard SOT-23 / 0805 / 1206 footprint.